

EMG4 exploratory analyses

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Overview

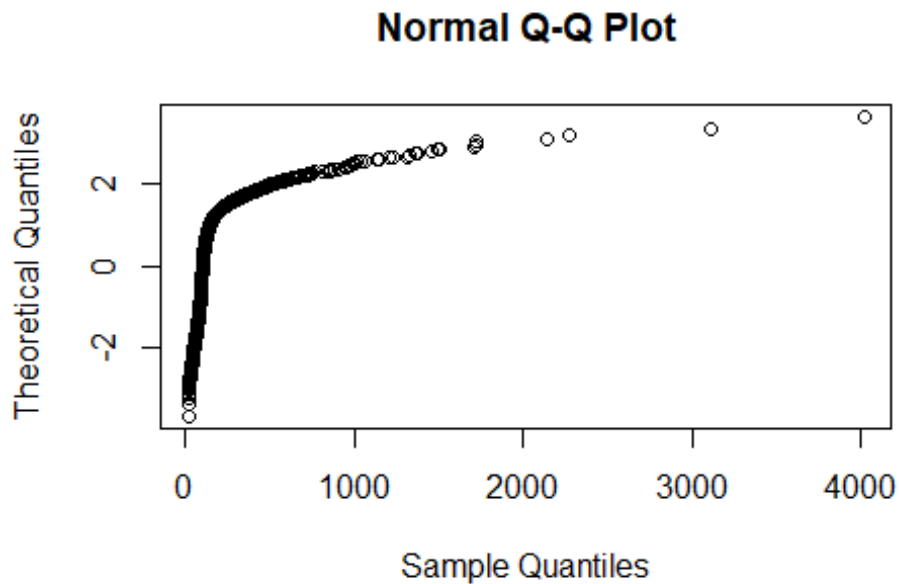
This document contains the exploratory statistical analyses of the EMG4 data combined with the embedded ratings, justice and fairness scores.

Load preprocessed MLM EMG data

1. Embedded Ratings

For exploratory reasons, we also look at the correlation between the EMG signal at the character morality segment and the ratings on the embedded rating task. We focus on the peak of the differential effect, which occurs 3-4s after the onset of the segment. The higher the rating score, the more moral the behavior was judged. So we expect to find a negative correlation, more frowning to immoral behavior (i.e. lower scores).

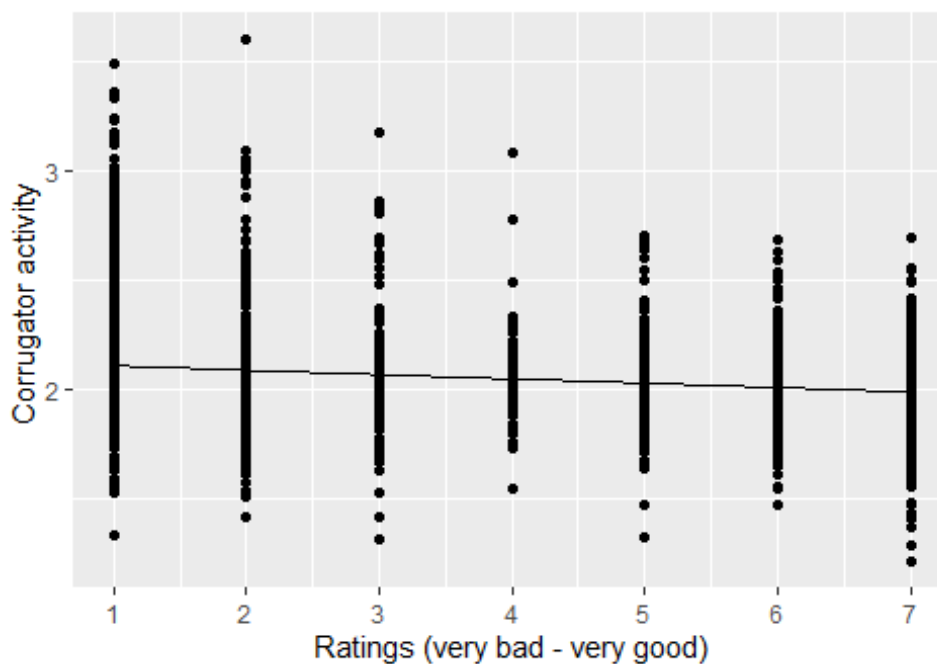
Check the normal distribution of the morality responses



Q-Q plot to check normality om EMG results at the Morality segment.

The averaged EMG data are not normally distributed, so we take the log transform.

Plot the relation between centered embedded rating and log corrugator EMG at the peak at the Morality Segment



Scatterplot between the centered embedded rating and log corrugator EMG at the peak at the Morality Segment.

Pearson's product-moment correlation: embedded ratings and morality EMG signal

```
##
## Pearson's product-moment correlation
##
## data: embed_rating and Morality_mean_log
## t = -14.175, df = 3699, p-value < 2.2e-16
## alternative hypothesis: true correlation is not equal to 0
## 95 percent confidence interval:
## -0.2573231 -0.1962015
## sample estimates:
## cor
## -0.2269858
```

There is a weak negative correlation: $r(3699) = -0.2269858$, $p = 1.8672345 \times 10^{-44}$.

As expected, the lower the rating, the higher the corrugator EMG activation. There were some NA ratings and some trials were excluded in the baseline procedure. In total there was a data loss of 3.6197917%.

2. Justice score

2.0 Load final MLM models from the regular analysis

2.1 Justice score and EMG signal at morality segment

The best model was Morality12: $\text{Morality_response} \sim \text{MoralityCondition} + \text{t1_immoral} + \text{t1_moral} + \text{t2_immoral} + \text{t3_immoral} + (1 + \text{MoralityCondition} \mid \text{sjCode}) + (1 \mid \text{Markertrial})$. We now add our justice score to that model and check whether it improves the fit.

```
load(paste0(getwd(), "/WORCS_EMG4/results/EMG4_MoralityJustice_final_model_60sjs.RData"))
load(paste0(getwd(), "/WORCS_EMG4/results/EMG4_Morality_js_model_60sjs.RData"))

anova(MoralityM12_js, MoralityM12_justice_score) # p = .068, so MoralityM12_js is the better model

## Data: justice_mor_data
## Models:
## MoralityM12_js: Morality_response ~ MoralityCondition + t1_immoral + t1_moral + t2_immoral +
t3_immoral + (1 + MoralityCondition | sjCode) + (1 | Markertrial)
## MoralityM12_justice_score: Morality_response ~ MoralityCondition * justice_score + t1_immoral +
t1_moral + t2_immoral + t3_immoral + (1 + MoralityCondition | sjCode) + (1 | Markertrial)
## npar AIC BIC logLik deviance Chisq Df
## MoralityM12_js 11 2348055 2348166 -1174017 2348033
## MoralityM12_justice_score 13 2348054 2348185 -1174014 2348028 5.3626 2
## Pr(>Chisq)
## MoralityM12_js
## MoralityM12_justice_score 0.06848 .
## ---
## Signif. codes: 0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
```

Adding the justice score to the MoralityM12 model does not improve the fit of the model ($p = .068$).

2.2 Justice score and EMG signal at state adjective segment

The best model is Adjective17: $\text{StateAdjective_response} \sim \text{MoralityCondition} + \text{StateAdjectiveCondition} + \text{MoralityCondition}:\text{StateAdjectiveCondition} + \text{t1_moral_neg} + \text{t1_moral_pos} + \text{t2_immoral_neg} + \text{t2_moral_neg} + (1 + \text{t1_moral_neg} + \text{t2_immoral_neg} \mid \text{sjCode}) + (1 \mid \text{Markertrial})$.

```
## Data: justice_adj_data
## Models:
## AdjectiveM17_js: StateAdjective_response ~ MoralityCondition * StateAdjectiveCondition +
t1_moral_neg + t1_moral_pos + t2_immoral_neg + t2_moral_neg + (1 + t1_moral_neg + t2_immoral
```

```

_neg |
sjCode) + (1 | Markertrial)
## AdjectiveM17_justice_score_full: StateAdjective_response ~ MoralityCondition *
StateAdjectiveCondition * justice_score + t1_moral_neg + t1_moral_pos + t2_immoral_neg +
t2_moral_neg + (1 + t1_moral_neg + t2_immoral_neg | sjCode) + (1 | Markertrial)
## npar AIC BIC logLik deviance Chisq Df
## AdjectiveM17_js 16 655231 655374 -327600 655199
## AdjectiveM17_justice_score_full 20 655194 655372 -327577 655154 45.811 4
## Pr(>Chisq)
## AdjectiveM17_js
## AdjectiveM17_justice_score_full 2.697e-09 ***
## ---
## Signif. codes: 0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1

```

Adding the justice score to the full factorial model improves the fit.

Table 2 State Adjective Segment Linear Mixed Models Justice Analysis Summary.

<i>Predictors</i>	State Adjective Segment (M17_full, <i>df</i> = 55165)		
	<i>Estimates</i> (95% <i>CI</i>)	<i>z</i>	<i>p</i>
(Intercept)	109.97 (91.67, 128.26)	11.78	<0.001*
MoralityCondition [Immoral]	0.31 (-3.47, 4.08)	0.16	0.87
StateAdjectiveCondition [Negative]	12.69 (8.52, 16.85)	5.97	<0.001*
Justice score	1.31 (-22.56, 25.18)	0.11	0.92
t1 moral negative	8.44 (-3.42, 20.29)	1.39	0.17
t1 moral positive	-5.51 (-9.04, -1.99)	-3.07	0.002*
t2 immoral negative	-11.76 (-35.46, 11.93)	-0.97	0.33
t2 moral negative	-35.33 (-44.47, -26.19)	-7.58	<0.001*
MoralityCondition [Immoral] * StateAdjectiveCondition [Negative]	0.76 (-5.52, 7.05)	0.24	0.81
MoralityCondition [Immoral] * Justice score	10.66 (5.46, 15.85)	4.02	<0.001*
StateAdjectiveCondition [Negative] * Justice score	17.66 (12.45, 22.86)	6.65	<0.001*
MoralityCondition [Immoral] * StateAdjectiveCondition [Negative] * Justice score	-20.93 (-29.18, -12.68)	-4.97	<0.001*
Random Effects			
σ^2	8272.36		
τ_{00} Markertrial	117.18		
τ_{00} sjCode	1821.48		
τ_{11} sjCode.t1_moral_negative	1967.61		
τ_{11} sjCode.t2_immoral_negative	7330.35		
ρ_{01} sjCode.t1_moral_negative	0.17		
ρ_{01} sjCode.t2_immoral_negative	-0.32		
ICC	0.20		
Deviance	655153.66		
log-Likelihood	-327576.83		

Note. Best model for the analysis of corrugator activity ($N_{\text{sjCode}} = 59$, $N_{\text{Markertrial}} = 64$). * significant *p*-value.

2.2.1 Probing the 3-way interaction

We can probe the 3-way interaction, provided we choose specific values for the justice scores. The scores range from -0.2941176 to 1.4705882, since -1.2353 was an outlier we choose the values from -0.5 till 1.5 with increments of 0.5.

Table 3 Estimated Marginal Means for the State Adjective Segment at different levels of the Justice score.

StateAdjectiveCondition	MoralityCondition	justice_score	emmean	df	lower.CL	upper.CL
Positive	Moral	-0.5	107.108	64.971	78.030	136.187
Negative	Moral	-0.5	110.965	65.179	81.865	140.065
Positive	Immoral	-0.5	102.086	64.956	73.009	131.163
Negative	Immoral	-0.5	117.173	60.928	88.543	145.802
Positive	Moral	0.0	107.762	68.765	89.302	126.221
Negative	Moral	0.0	120.447	69.357	101.951	138.943
Positive	Immoral	0.0	108.068	68.745	89.610	126.526
Negative	Immoral	0.0	121.516	67.047	103.247	139.785
Positive	Moral	0.5	108.415	68.697	96.889	119.941
Negative	Moral	0.5	129.929	70.246	118.343	141.514
Positive	Immoral	0.5	114.049	68.680	102.524	125.575
Negative	Immoral	0.5	125.859	70.280	114.285	137.434
Positive	Moral	1.0	109.068	70.546	94.239	123.897
Negative	Moral	1.0	139.411	71.463	124.537	154.284
Positive	Immoral	1.0	120.031	70.543	105.202	134.860
Negative	Immoral	1.0	130.203	70.555	115.455	144.950
Positive	Moral	1.5	109.722	66.169	85.158	134.285
Negative	Moral	1.5	148.893	66.450	124.305	173.480
Positive	Immoral	1.5	126.013	66.166	101.449	150.576
Negative	Immoral	1.5	134.546	62.731	110.331	158.760

Table 4 Pairwise comparison between Negative and Positive State Adjectives at different levels of the Justice score.

contrast	justice_score	MoralityCondition	estimate	df	lower.CL	upper.CL	t.ratio	p.value
Negative - Positive	-0.5	Moral	3.857	55012.10	-2.503	10.216	1.189	0.235
Negative - Positive	-0.5	Immoral	15.087	12663.08	7.535	22.638	3.916	0.000
Negative - Positive	0.0	Moral	12.685	55027.71	8.519	16.852	5.967	0.000
Negative - Positive	0.0	Immoral	13.448	18732.07	8.725	18.171	5.581	0.000
Negative - Positive	0.5	Moral	21.514	54974.93	18.716	24.312	15.071	0.000
Negative - Positive	0.5	Immoral	11.810	54634.75	8.990	14.629	8.210	0.000
Negative - Positive	1.0	Moral	30.342	55019.29	26.900	33.785	17.274	0.000

contrast	justice_score	MoralityCondition	estimate	df	lower.CL	upper.CL	t.ratio	p.value
Negative - Positive	1.0	Immoral	10.172	29102.91	6.436	13.907	5.337	0.000
Negative - Positive	1.5	Moral	39.171	55023.32	33.746	44.596	14.152	0.000
Negative - Positive	1.5	Immoral	8.533	14147.81	2.183	14.883	2.634	0.008

We can also probe the 3-way interaction by inspecting the trends for the justice scores in the different conditions.

Table 5 Estimated Marginal Trends for the State Adjective Segment at different levels of the Justice score.

StateAdjectiveCondition	MoralityCondition	justice_score.trend	df	lower.CL	upper.CL
Positive	Moral	1.307	61.598	-23.040	25.654
Negative	Moral	18.964	61.562	-5.380	43.308
Positive	Immoral	11.963	61.590	-12.383	36.310
Negative	Immoral	8.687	55.999	-15.202	32.575

Table 6 Pairwise comparison between Negative and Positive State Adjectives at different levels of the Morality condition.

contrast	estimate	df	lower.CL	upper.CL	t.ratio	p.value
Positive Moral - Negative Moral	-17.657	54987.80	-24.647	-10.667	-6.647	0.000
Positive Moral - Positive Immoral	-10.656	55027.08	-17.630	-3.683	-4.021	0.000
Positive Moral - Negative Immoral	-7.380	10044.77	-15.913	1.153	-2.276	0.130
Negative Moral - Positive Immoral	7.001	55020.54	0.065	13.936	2.656	0.047
Negative Moral - Negative Immoral	10.277	10001.22	1.751	18.804	3.172	0.009
Positive Immoral - Negative Immoral	3.277	10127.01	-5.291	11.844	1.006	0.896

2.2.2 Plotting the 3-way interaction

Plot of regression lines for the different categories of Morality and Valence.

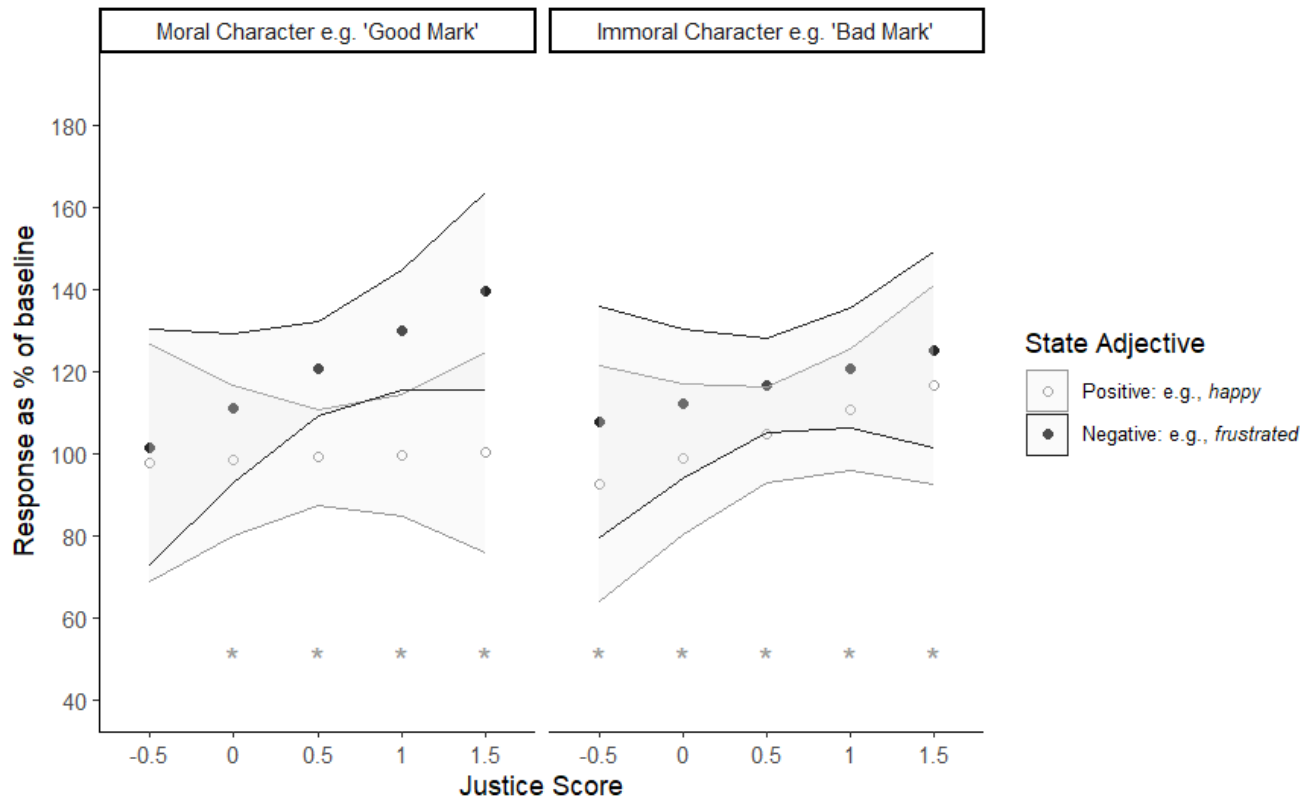


Figure on State Adjective segment for different justice scores

People with high vs, low justice trait scores have different responses in the moral conditions compared to the immoral conditions, as was indicated by the significant 3-way interaction (see Table 2, $b = -20.93$, $z = -4.97$, $p < 0.001$).

In the moral condition the participants who scored low on the justice trait score (i.e. people who do not agree to want positive things for good people and negative things for bad people) do not show a differential response towards the positive and negative state adjectives. Participants who were indifferent, or agreed to want positive things for good people and negative things for bad people, i.e. have a higher justice score, frowned more in response to negative state adjectives compared to positive state adjectives. As per our multiple-drivers model frowning at negative state adjectives for moral characters could be the net result of negative simulation and additional negative evaluation (unfair). Our justice trait score seems to influence the degree of frowning in this case, hinting at a possible attenuation of the negative evaluation. Only participants who value a 'sense of justice' frown at unfair events that befall a moral character.

Interestingly, in the immoral conditions all participants, irrespective of their justice score, frowned more in response to negative state adjectives than to positive state adjectives. This pattern can be explained by negative simulation rather than evaluation, as the evaluation would counteract the simulation force yielding no differential responses or even a flipped response pattern. This is clearly not the case. However, the higher the justice score, the higher the overall response was (see Table 2, $b = 10.66$, $z = 4.02$, $p < 0.001$).

2.3 Justice score and EMG signal at affect reason segment

The best model is Reason22: AffectReason_response ~ MoralityCondition + StateAdjectiveCondition + MoralityCondition:StateAdjectiveCondition + t1_moral_neg + t1_moral_pos + + t2_moral_neg + t3_moral_neg + (1 + t1_moral_neg + t3_moral_neg | sjCode) + (1 | Markertrial).

```
## Data: justice_reason_data
## Models:
## ReasonM22_js: AffectReason_response ~ MoralityCondition * StateAdjectiveCondition + t1_moral_neg
+ t1_moral_pos + t2_moral_neg + t3_moral_neg + (1 + t1_moral_neg + t3_moral_neg | sjCode) +
(1 | Markertrial)
## ReasonM22_justice_score_alt: AffectReason_response ~ MoralityCondition * StateAdjectiveCondition
+ justice_score + t1_moral_neg + t1_moral_pos + t2_moral_neg + t3_moral_neg + (1 + t1_moral_neg |
sjCode) + (1 | Markertrial)
## npar AIC BIC logLik deviance Chisq Df
## ReasonM22_js 16 1119977 1120128 -559972 1119945
## ReasonM22_justice_score_alt 17 1119837 1119997 -559901 1119803 141.94 1
## Pr(>Chisq)
## ReasonM22_js
## ReasonM22_justice_score_alt < 2.2e-16 ***
## ---
## Signif. codes: 0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
```

Adding the justice score to the full factorial model improves the fit.

Table 7 Affect Reason Segment Linear Mixed Models Justice Analysis Summary.

<i>Predictors</i>	Affect Reason Segment (M22_alt, df = 91958)		
	<i>Estimates</i> (95% CI)	<i>z</i>	<i>p</i>
(Intercept)	102.90 (85.73, 120.06)	11.75	<0.001*
MoralityCondition [Immoral]	1.50 (-1.92, 4.92)	0.86	0.39
StateAdjectiveCondition [Negative]	22.56 (18.80, 26.33)	11.75	<0.001*
Justice score	2.91 (-20.17, 26.00)	0.25	0.81
t1 moral negative	6.77 (1.45, 12.10)	2.49	0.013*
t1 moral positive	-3.87 (-5.78, -1.96)	-3.97	<0.001*
t2 moral negative	-14.79 (-17.74, -11.83)	-9.81	<0.001*
t3 moral negative	-4.86 (-9.55, -0.16)	-2.03	0.043*
MoralityCondition [Immoral] * StateAdjectiveCondition [Negative]	-10.85 (-15.93, -5.76)	-4.18	<0.001*
MoralityCondition [Immoral] * Justice score	20.75 (16.05, 25.45)	8.66	<0.001*
StateAdjectiveCondition [Negative] * Justice score	28.72 (24.01, 33.42)	11.96	<0.001*
MoralityCondition [Immoral] * StateAdjectiveCondition [Negative] * Justice score	-42.13 (-48.80, -35.47)	-12.39	<0.001*
Random Effects			
σ^2	11284.21		
T00 Markertrial	155.56		
T00 sjCode	1386.22		
T11 sjCode.t1_moral_negative	84.37		

ρ_{01} sjCode.t1_moral_negative	-0.01
ICC	0.12
Deviance	1119802.81
log-Likelihood	-559901.41

Note. Best model for the analysis of corrugator activity ($N_{\text{sjCode}} = 59$, $N_{\text{Markertrial}} = 64$). * significant p -value.

2.3.1 Probing the 3-way interaction

We can probe the 3-way interaction, provided we choose specific values for the justice scores.

Table 8 Estimated Marginal Means for the Affect Reason Segment at different levels of the Justice score.

StateAdjectiveCondition	MoralityCondition	justice_score	emmean	df	lower.CL	upper.CL
Positive	Moral	-0.5	99.51	62.27	71.602	127.42
Negative	Moral	-0.5	107.72	62.46	79.79	135.65
Positive	Immoral	-0.5	90.64	62.27	62.73	118.55
Negative	Immoral	-0.5	109.06	62.28	81.15	136.97
Positive	Moral	0.0	100.97	64.75	83.47	118.47
Negative	Moral	0.0	123.53	65.26	106.00	141.06
Positive	Immoral	0.0	102.47	64.74	84.97	119.97
Negative	Immoral	0.0	114.19	64.75	96.69	131.69
Positive	Moral	0.5	102.43	72.37	91.90	112.96
Negative	Moral	0.5	139.35	73.95	128.76	149.93
Positive	Immoral	0.5	114.30	72.35	103.77	124.83
Negative	Immoral	0.5	119.31	72.36	108.78	129.84
Positive	Moral	1.0	103.88	67.24	89.99	117.77
Negative	Moral	1.0	155.16	68.05	141.23	169.09
Positive	Immoral	1.0	126.13	67.23	112.25	140.02
Negative	Immoral	1.0	124.43	67.18	110.55	138.32
Positive	Moral	1.5	105.34	62.95	81.84	128.84
Negative	Moral	1.5	170.98	63.18	147.46	194.50
Positive	Immoral	1.5	137.97	62.94	114.47	161.47
Negative	Immoral	1.5	129.56	62.90	106.06	153.05

Table 9 Pairwise comparison between Negative and Positive State Adjectives at different levels of the Justice score.

contrast	justice_score	MoralityCondition	estimate	df	lower.CL	upper.CL	t.ratio	p.value
Negative - Positive	-0.5	Moral	8.20	91850.29	2.46	13.95	2.80	0.005
Negative - Positive	-0.5	Immoral	18.43	91841.94	12.89	23.96	6.53	0.000
Negative - Positive	0.0	Moral	22.56	91857.55	18.80	26.33	11.75	0.000
Negative - Positive	0.0	Immoral	11.72	91854.04	8.28	15.15	6.69	0.000
Negative - Positive	0.5	Moral	36.92	91821.82	34.39	39.45	28.64	0.000
Negative - Positive	0.5	Immoral	5.01	91831.13	3.01	7.01	4.90	0.000

contrast	justice_score	MoralityCondition	estimate	df	lower.CL	upper.CL	t.ratio	p.value
Negative - Positive	1.0	Moral	51.28	91849.83	48.17	54.39	32.32	0.000
Negative - Positive	1.0	Immoral	-1.70	91856.73	-4.40	1.00	-1.24	0.217
Negative - Positive	1.5	Moral	65.64	91855.71	60.74	70.54	26.25	0.000
Negative - Positive	1.5	Immoral	-8.41	91849.36	-13.06	-3.76	-3.55	0.000

We can also probe the 3-way interaction by inspecting the trends for the justice scores in the different conditions.

Table 10 Estimated Marginal Trends for the Affect Reason Segment at different levels of the Justice score.

StateAdjectiveCondition	MoralityCondition	justice_score.trend	df	lower.CL	upper.CL
Positive	Moral	2.91	60.74	-20.64	26.47
Negative	Moral	31.63	60.71	8.08	55.18
Positive	Immoral	23.66	60.73	0.11	47.22
Negative	Immoral	10.25	60.718	-13.31	33.80

Table 11 Pairwise comparison between Negative and Positive State Adjectives at different levels of the Morality condition.

contrast	estimate	df	lower.CL	upper.CL	t.ratio	p.value
Positive Moral - Negative Moral	-28.72	91837.37	-35.03	-22.40	-11.96	0.000
Positive Moral - Positive Immoral	-20.75	91857.65	-27.06	-14.45	-8.66	0.000
Positive Moral - Negative Immoral	-7.33	91850.94	-13.61	-1.06	-3.07	0.013
Negative Moral - Positive Immoral	7.97	91850.98	1.70	14.24	3.34	0.005
Negative Moral - Negative Immoral	21.38	91857.37	15.11	27.65	8.97	0.000
Positive Immoral - Negative Immoral	13.42	91831.81	7.10	19.73	5.59	0.000

2.3.2 Plotting the 3-way interaction

Plot of regression lines for the different categories of Morality and Valence.

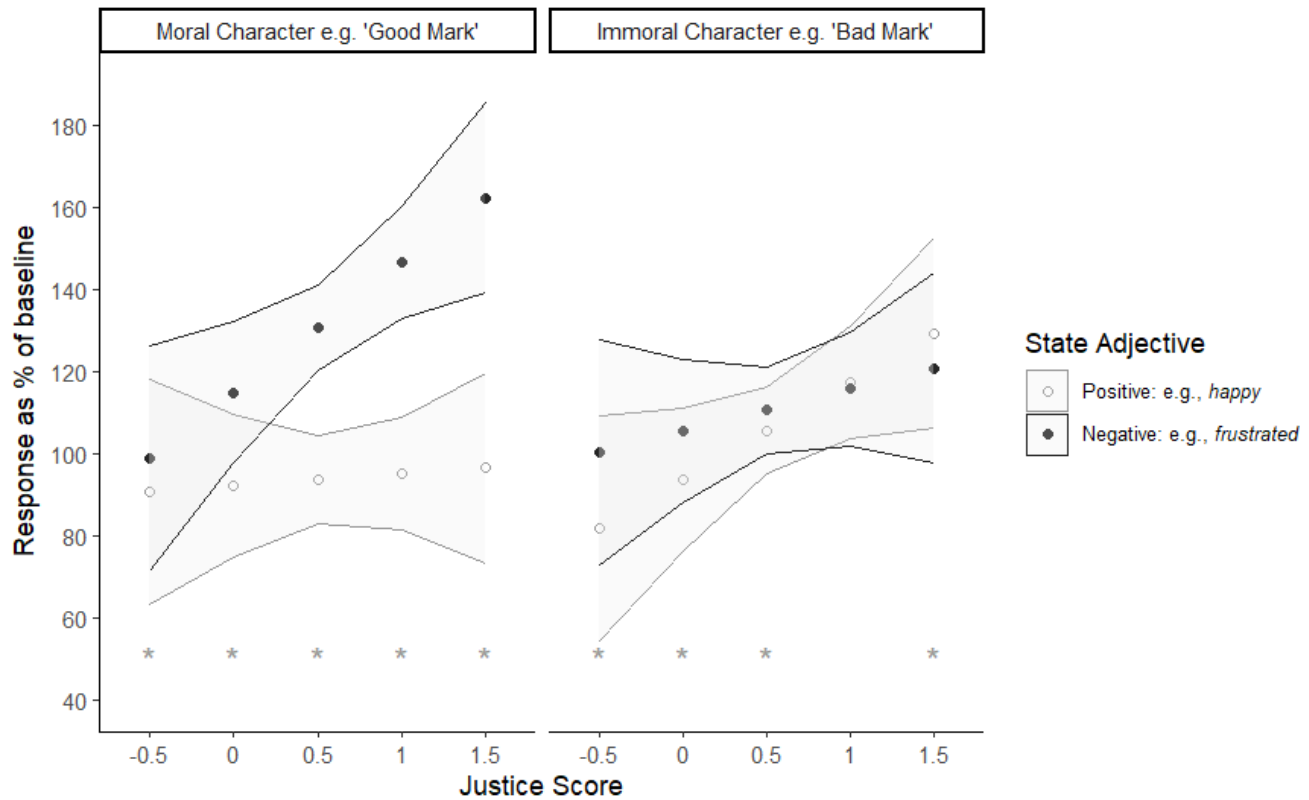


Figure on Affect Reason segment for different justice scores

People with high vs, low justice trait scores have different responses in the moral conditions than in the immoral conditions, as was indicated by the significant 3-way interaction (see Table 7, $b = -42.13$, $z = -12.39$, $p < 0.001$).

In the moral condition, similar to the responses at the state adjective, participants frowned more in response to negative state adjectives compared to positive state adjectives and this difference increases as their justice score increases. As per our multiple-drivers model frowning at negative state adjectives for moral characters could be the net result of negative simulation and additional negative evaluation (unfair). Our justice trait score seems to influence the degree of frowning in this case, hinting at a possible attenuation of the negative evaluation. Participants who stronger value a 'sense of justice' frown substantially more at unfair events that befall a moral character.

Interestingly, in the immoral conditions participants, with a low to moderate justice score, frowned more in response to negative state adjectives than to positive state adjectives. This pattern can be explained by negative simulation rather than evaluation, as the evaluation would counteract the simulation force yielding no differential responses or even a flipped response pattern. This is clearly not the case. However, for participants with a higher justice score the pattern starts to flip. For the participants with the highest justice score we see that they frown more in response to positive state adjectives than negative state adjectives (see Table 9, $b = -8.41$, $t = -3.56$, $p < 0.001$). This is the pattern that we had hoped to find overall and would indicate a counteracting force of fair evaluation and negative simulation of negative events befalling an immoral character.

3. Fairness ratings

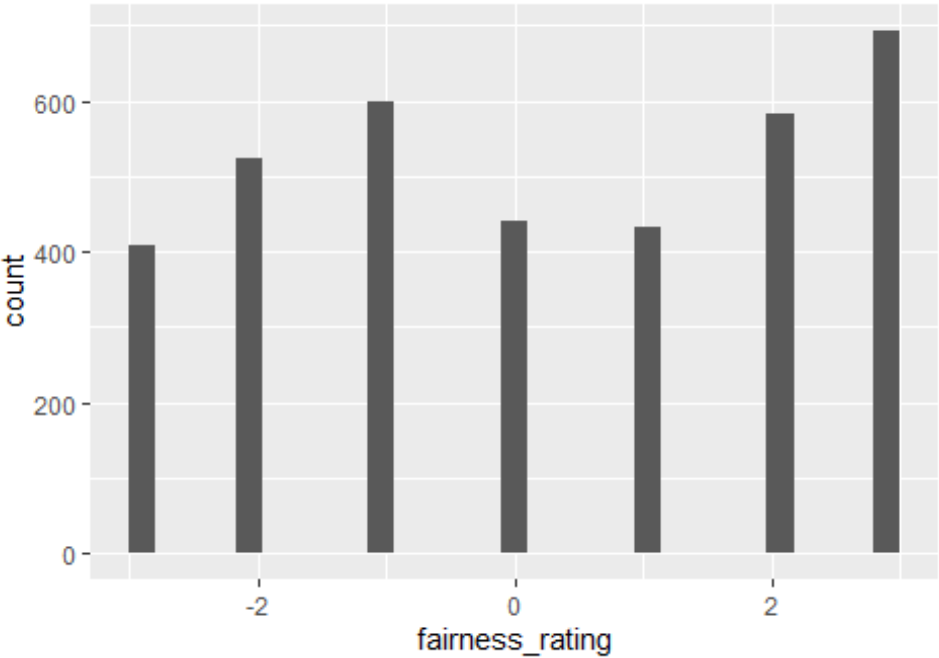
With the Story Fairness rating responses we were able to explore if and how the hypothesized effect of fairness-based evaluation would be reflected in a self-report measure as evaluation for this response tends to occur on a more conscious level. Therefore, we built a linear mixed model again focused on the interaction between the fixed factors morality condition (moral, immoral) and event valence (positive, negative) and with subject and item as random effects, but now with the Story Fairness ratings as the outcome measure. We also performed pairwise comparisons to investigate significant effects.

A combined amount of 87 ratings from 7 different participants were excluded from the analysis due to an error in the creation of the four versions of the Story Fairness Tasks (corresponding to the four versions of stimuli lists). These issues were quickly fixed so the data of all other participants was complete. Furthermore, there was 1 participant whose ratings on all 64 narratives were excluded from analysis as everything was rated as neutral. This resulted in a total data loss of 4.0625%.

A summary of the distribution and the means of the 4 conditions:

```
##
##  -3  -2  -1   0   1   2   3
## 408 525 599 442 434 583 694

##
##          -1.4375 -1.33224755700326  1.61195652173913  2.05901639344262
##          1          1          1          1
```



Histogram of Fairness ratings

3.1 Building the Fairness rating model

Table 12 Fairness Ratings Linear Mixed Models Analysis Summary.

	Fairness Ratings
	(M1, df = 3677)

<i>Predictors</i>	<i>Estimates (95% CI)</i>	<i>z</i>	<i>p</i>
(Intercept)	2.06 (1.97, 2.16)	42.42	<0.001*
MoralityCondition [Immoral]	-3.39 (-3.50, -3.28)	-60.08	<0.001*
StateAdjectiveCondition [Negative]	-3.50 (-3.61, -3.39)	-62.11	<0.001*
MoralityCondition [Immoral] * StateAdjectiveCondition [Negative]	6.44 (6.29, 6.60)	80.85	<0.001*
Random Effects			
σ^2	1.46		
τ_{00} Markertrial	0.02		
τ_{00} sjCode	0.02		
ICC	0.03		
Deviance	11932.39		
log-Likelihood	-5966.19		

Note. Best model for the analysis of corrugator activity ($N_{\text{sjCode}} = 59$, $N_{\text{Markertrial}} = 64$). * significant p -value.

3.1.1 Post-hoc analyses

Table 13 Estimated Marginal Means for the Fairness ratings at different Morality conditions and State Adjectives.

MoralityCondition	StateAdjectiveCondition	emmean	df	lower.CL	upper.CL
Moral	Positive	2.06	277.38	1.97	2.16
Immoral	Positive	-1.33	275.36	-1.43	-1.24
Moral	Negative	-1.44	272.82	-1.53	-1.34
Immoral	Negative	1.61	275.61	1.52	1.71

Table 14 Pairwise comparisons for the Fairness ratings at different Morality conditions .

MoralityCondition	contrast	estimate	df	lower.CL	upper.CL	t.ratio	p.value
Moral	Negative - Positive	-3.50	3568.25	-3.64	-3.36	-62.11	<0.001*
Immoral	Negative - Positive	2.94	3566.56	2.80	3.08	52.24	<0.001*

Plot of Fairness ratings for the different State Adjectives for the Morality manipulations:



Figure on Fairness Ratings

3.2 Fairness rating and Justice Score

The model for Fairness was FairnessM1: $\text{fairness_rating} \sim \text{MoralityCondition} * \text{StateAdjectiveCondition} + (1|\text{sjCode}) + (1|\text{Markertrial})$. We now add our justice score to that model and check whether it improves the fit.

```
## Data: justice_fairness_data
## Models:
## FairnessM1_justice: fairness_rating ~ MoralityCondition * StateAdjectiveCondition + (1 |
sjCode)
+ (1 | Markertrial)
## FairnessM1_justice_full: fairness_rating ~ MoralityCondition * StateAdjectiveCondition *
justice_score + (1 | sjCode) + (1 | Markertrial)
## npar AIC BIC logLik deviance Chisq Df Pr(>Chisq)
## FairnessM1_justice 7 11946 11990 -5966.2 11932
## FairnessM1_justice_full 11 11884 11952 -5931.0 11862 70.46 4 1.815e-14
##
## FairnessM1_justice
## FairnessM1_justice_full ***
## ---
## Signif. codes: 0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
```

Adding the justice score to the full factorial model improves the fit.

Table 15 Fairness ratings Linear Mixed Models Justice Analysis Summary.

Predictors	Fairness Scores (M1_full, df = 3673)		
	Estimates (95% CI)	z	p
(Intercept)	1.95 (1.81, 2.09)	27.03	<0.001*
MoralityCondition [Immoral]	-3.01 (-3.19, -2.84)	-34.28	<0.001*
StateAdjectiveCondition [Negative]	-3.25 (-3.42, -3.07)	-36.99	<0.001*

Justice score	0.19 (0.01, 0.37)	2.06	0.039*
MoralityCondition [Immoral] *	5.67		
StateAdjectiveCondition [Negative]	(5.43, 5.92)	45.57	<0.001*
MoralityCondition [Immoral] *	-0.65		
Justice score	(-0.88, -0.42)	-5.57	<0.001*
StateAdjectiveCondition [Negative]	-0.44		
* Justice score	(-0.67, -0.21)	-3.72	<0.001*
MoralityCondition [Immoral] *	1.33		
StateAdjectiveCondition [Negative]	(1.00, 1.65)	7.97	<0.001*
* Justice score			
Random Effects			
σ^2	1.43		
τ_{00} Markertrial	0.02		
τ_{00} sjCode	0.02		
ICC	0.03		
Deviance	11861.93		
log-Likelihood	-5930.96		

Note. Best model for the analysis of corrugator activity ($N_{sjCode} = 59$, $N_{Markertrial} = 64$). * significant p -value.

3.2.1 Probing the 3-way interaction

We can probe the 3-way interaction, provided we choose specific values for the justice scores.

Table 16 Estimated Marginal Means for the Fairness ratings at different levels of the Justice score.

StateAdjectiveCondition	MoralityCondition	justice_score	emmean	df	lower.CL	upper.CL
Positive	Moral	-0.5	1.85	351.13	1.63	2.07
Negative	Moral	-0.5	-1.18	348.65	-1.39	-0.96
Positive	Immoral	-0.5	-0.83	351.12	-1.05	-0.61
Negative	Immoral	-0.5	1.15	349.91	0.93	1.37
Positive	Moral	0.0	1.95	347.92	1.81	2.09
Negative	Moral	0.0	-1.30	344.61	-1.44	-1.16
Positive	Immoral	0.0	-1.06	347.53	-1.21	-0.92
Negative	Immoral	0.0	1.37	346.33	1.22	1.51
Positive	Moral	0.5	2.05	281.68	1.95	2.14
Negative	Moral	0.5	-1.42	277.16	-1.51	-1.32
Positive	Immoral	0.5	-1.29	279.86	-1.39	-1.20
Negative	Immoral	0.5	1.58	279.85	1.48	1.67
Positive	Moral	1.0	2.14	336.20	2.02	2.26
Negative	Moral	1.0	-1.54	332.19	-1.66	-1.42
Positive	Immoral	1.0	-1.53	334.20	-1.65	-1.40
Negative	Immoral	1.0	1.79	334.89	1.67	1.92
Positive	Moral	1.5	2.24	353.60	2.04	2.43
Negative	Moral	1.5	-1.66	350.84	-1.86	-1.47
Positive	Immoral	1.5	-1.76	352.51	-1.95	-1.56
Negative	Immoral	1.5	2.01	352.72	1.82	2.20

Table 17 Pairwise comparison between Negative and Positive State Adjectives at different levels of the Justice score.

contrast	justice_score	MoralityCondition	estimate	df	lower.CL	upper.CL	t.ratio	p.value
Negative - Positive	-0.5	Moral	-3.03	3619.10	-3.30	-2.76	-21.94	<.001*
Negative - Positive	-0.5	Immoral	1.98	3618.45	1.71	2.25	14.36	<.001*
Negative - Positive	0.0	Moral	-3.25	3609.93	-3.42	-3.07	-36.99	<.001*
Negative - Positive	0.0	Immoral	2.43	3608.87	2.26	2.60	27.66	<.001*
Negative - Positive	0.5	Moral	-3.46	3571.35	-3.58	-3.35	-61.25	<.001*
Negative - Positive	0.5	Immoral	2.87	3569.57	2.76	2.98	50.78	<.001*
Negative - Positive	1.0	Moral	-3.68	3598.64	-3.83	-3.54	-49.44	<.001*
Negative - Positive	1.0	Immoral	3.32	3597.83	3.17	3.46	44.54	<.001*
Negative - Positive	1.5	Moral	-3.90	3616.01	-4.14	-3.66	-32.12	<.001*
Negative - Positive	1.5	Immoral	3.76	3615.51	3.53	4.00	30.98	<.001*

We can also probe the 3-way interaction by inspecting the trends for the justice scores in the different conditions.

Table 18 Estimated Marginal Trends for the Fairness ratings.

StateAdjectiveCondition	MoralityCondition	justice_score.trend	df	lower.CL	upper.CL	t.ratio	p.value
Positive	Moral	0.19	339.40	0.01	0.38	2.06	0.040
Negative	Moral	-0.24	337.60	-0.43	-0.06	-2.62	0.009
Positive	Immoral	-0.46	339.33	-0.64	-0.28	-4.95	<0.001
Negative	Immoral	0.43	338.65	0.25	0.61	4.61	<0.001

Table 19 Pairwise comparison between Negative and Positive State Adjectives at different levels of the Morality condition.

contrast	estimate	df	lower.CL	upper.CL	t.ratio	p.value
Positive Moral - Negative Moral	0.44	3622.90	0.13	0.74	3.72	0.001
Positive Moral - Positive Immoral	0.65	3622.69	0.35	0.96	5.57	<0.001
Positive Moral - Negative Immoral	-0.24	3566.42	-0.54	0.07	-2.03	0.227
Negative Moral - Positive Immoral	0.22	3566.27	-0.09	0.52	1.87	0.320
Negative Moral - Negative Immoral	-0.67	3622.89	-0.98	-0.37	-5.75	<0.001
Positive Immoral - Negative Immoral	-0.89	3622.53	-1.20	-0.58	-7.60	<0.001

3.2.2 Plotting the 3-way interaction

Plot of regression lines for the different categories of Morality and Valence.

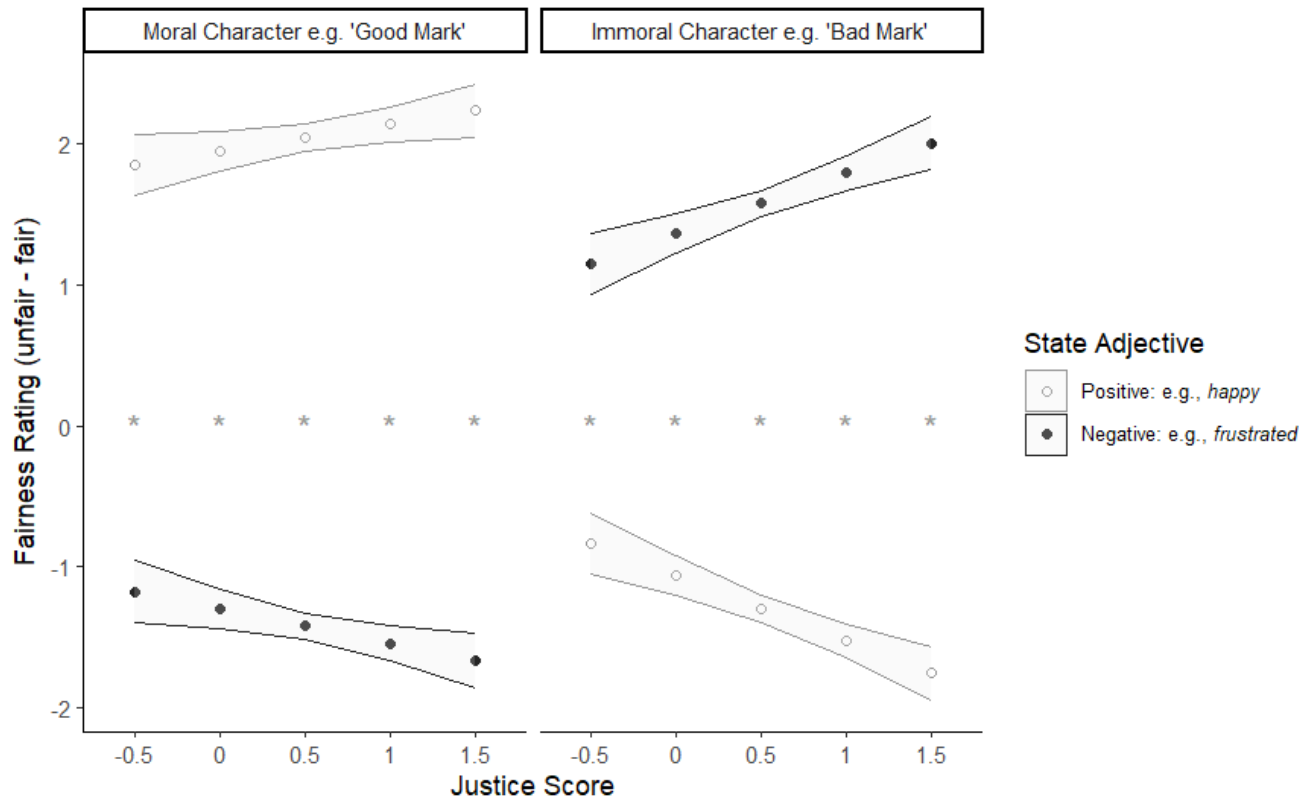


Figure on Fairness Ratings for different justice scores

People with high vs, low justice trait scores have different responses in the moral conditions than in the immoral conditions, as was indicated by the significant 3-way interaction (see Table 15, $b = 1.33$, $z = 7.97$, $p < 0.001$).

The patterns for moral and immoral conditions are similar with respect to the influence of justice score. Participants who stronger value a 'sense of justice' gave more extreme ratings, i.e. stronger fair judgments in fair situations and stronger unfair judgments in unfair situation. So the fairness judgments are positively related to the justice scores. What seems to happen is that the differential rating between positive and negative affect reasons is larger for participants with a higher justice score.